

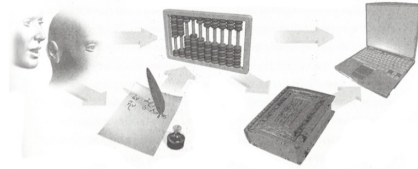
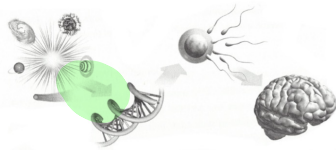
Natural Computing

LMU Munich
summer term 2025

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Artificial Chemistries



“The history of the universe can be thought of as a sequence of information processing revolutions, each of which builds on the technology of the previous ones.”

Seth Lloyd. *Programming the Universe*. Vintage Books, 2006.

On Multi-Sets...

Let X be a set with members from a space $\mathcal{X}$.

We can identify a set X with its membership function $\mu_X : \mathcal{X} \rightarrow \mathbb{B}$ such that

$$\mu_X(x) = 1 \iff x \in X.$$

For a multiset X' we generalize the membership function so that

$$\mu_{X'} : \mathcal{X} \rightarrow \mathbb{N}$$

$$\mu_{X'}(x) \geq 1 \iff x \in X'.$$

In both cases, the subset relation between (multi-)sets X, Y can be defined as

$$X \subseteq Y \iff \forall x \in X : \mu_X(x) \leq \mu_Y(x).$$

We then define $\wp(X) = 2^X = \mathbb{B}^X \cong X \rightarrow \mathbb{B}$,

and $\wp^*(X) = \mathbb{N}^X \cong X \rightarrow \mathbb{N}$,

to be the powerset functions for sets and multisets, respectively.

More on Multi-Sets...

example $\cap$:

define $X \cap Y$ given X, Y

$$\mu_{X \cap Y}(x) = \min \{ \mu_X(x), \mu_Y(x) \}$$

example $\cup$:

$$\mu_{X \cup Y}(x) = \max \{ \mu_X(x), \mu_Y(x) \}$$

$$\{3, 3, 4\} \cup \{2, 3, 4\} = \{2, 3, 3, 4\}$$

define $\oplus$:

$$\mu_{X \oplus Y}(x) = \mu_X(x) + \mu_Y(x)$$

$$\{3, 3, 4\} \oplus \{2, 3, 4\} =$$

$$= \{2, 3, 3, 3, 4, 4\}$$

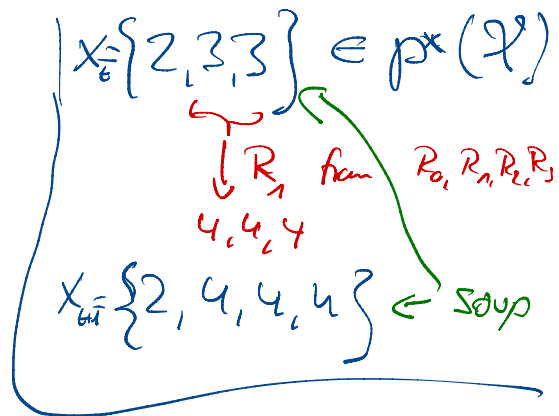
Definition 10 (artificial chemistry, soup). Let $\mathcal{X}$ be a state space. Let $\mathcal{R}$ be a set of reaction rules $R \in \mathcal{R}$ where each R is a function $R : \mathcal{X}_R^{\text{dom}} \rightarrow \mathcal{X}_R^{\text{cod}}$ for $\mathcal{X}_R^{\text{dom}}, \mathcal{X}_R^{\text{cod}} \subseteq \wp^*(\mathcal{X})$. Let $\mathcal{A}(X, \mathcal{R})$ be the set of all reaction materials from $X \in \wp^*(\mathcal{X})$ for all rules in $\mathcal{R}$, i.e.,

$$\mathcal{A}(X, \mathcal{R}) = \{(X', R) \mid X' \subseteq X \wedge X' \in \mathcal{X}_R^{\text{dom}} \wedge R \in \mathcal{R}\}.$$

Let $A : \wp^*(\mathcal{X}) \rightarrow \wp^*(\mathcal{X}) \times \mathcal{R}$ be a possibly randomized or non-deterministic function that takes a multi-set $X \in \wp^*(\mathcal{X})$ and returns a multi-set of reactants and a reaction rule to perform so that $A(X) \in \mathcal{A}(X, \mathcal{R})$. The tuple $(\mathcal{X}, \mathcal{R}, A)$ defines an artificial chemistry. A multi-set $X \in \wp^*(\mathcal{X})$ of an artificial chemistry is also called soup; each element $x \in X$ is also called a particle. The evolution of a soup for g generations is a sequence $\langle X_0, \dots, X_g \rangle$ so that X_0 is given as the initial soup and, for all t with $0 \leq t \leq g-1$

$$X_{t+1} = (X_t \setminus X') \uplus R(X')$$

where $(X', R) = A(X_t)$.



Definition 10 (artificial chemistry, soup). Let $\mathcal{X}$ be a state space. Let $\mathcal{R}$ be a set of reaction rules $R \in \mathcal{R}$ where each R is a function $R : \mathcal{X}_R^{\text{dom}} \rightarrow \mathcal{X}_R^{\text{cod}}$ for $\mathcal{X}_R^{\text{dom}}, \mathcal{X}_R^{\text{cod}} \subseteq \wp^*(\mathcal{X})$. Let $\mathcal{A}(X, \mathcal{R})$ be the set of all *reaction materials* from $X \in \wp^*(\mathcal{X})$ for all rules in $\mathcal{R}$, i.e.,

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$$X_{t+1} = (X_t \setminus X') \uplus R(X')$$

where $(X', R) = A(X_t)$.

Reaction rules are usually written in the form $R = \sum_i x_i \rightarrow \sum_j y_j$ for $x_i, y_j \in \mathcal{X}$, $i, j \in \mathbb{N}$, $\{x_i\}_i \in \mathcal{X}_R^{\text{dom}}$, $\{y_j\}_j \in \mathcal{X}_R^{\text{cod}}$, and $R(\{x_i\}_i) = \{y_j\}_j$.

Example from last slide:

$$R_1(X') =$$

$$\begin{cases} \{4, 4, 4\} & \text{if } X' = \{3, 3\} \\ \{5, 5, 5\} & \text{if } X' = \{4, 4\} \\ \text{undefined} & \text{otherwise} \end{cases}$$

$$R_1^0 = 3 + 3 \rightarrow 4 + 4 + 4$$

$$R_1^1 = 4 + 4 \rightarrow 5 + 5 + 5$$

$$\begin{aligned} 3 + 3 + 7 &\rightarrow 2 + 2 + 2 + 7 \\ 3 + 3 + 7 &\rightarrow 4 + 4 + 4 + 7 \end{aligned}$$

Example 1 ([2]). We imagine a colony of chameleons who can either assume a red, green, or blue color. Whenever two chameleons with a different color meet, they assume the third color (that none of them has).

Let $\mathcal{X} = \{r, g, b\}$.

Let $\mathcal{R} = \{R_1, R_2, R_3, R_4, R_5, R_6\}$ with

$$\rightarrow R_1 = r + r \rightarrow r + r,$$

$$\rightarrow R_2 = r + g \rightarrow b + b,$$

$$\rightarrow R_3 = r + b \rightarrow g + g,$$

$$\rightarrow R_4 = b + b \rightarrow b + b,$$

$$\rightarrow R_5 = b + g \rightarrow r + r,$$

$$\rightarrow \text{and } R_6 = g + g \rightarrow g + g.$$

$\{r, r, g, g\},$
 $\{r, b\}, \dots$

Let $A(X) \sim \mathcal{A}(X, \mathcal{R})$.

Let $X_0 \sim \wp^*(\mathcal{X})$.

or $X_0 = \{x_0, \dots, x_{n_0}\}$ where $x_i \sim \mathcal{X}$ for all i

Experiment Time!

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$$\text{and } R_6 = g + g \rightarrow g + g.$$

Let $A(X) \sim \mathcal{A}(X, \mathcal{R})$.

Let $X_0 \sim \wp^*(\mathcal{X})$.

early results



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late results



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Experiment Time! *again*

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$$R_3 = r + b \rightarrow g + g,$$

$$R_4 = b + b \rightarrow b + b,$$

$$R_5 = b + g \rightarrow r + r,$$

$$\text{and } R_6 = g + g \rightarrow g + g.$$

Let $A(X) \sim \mathcal{A}(X, \mathcal{R})$.

Let $X_0 \sim \wp^*(\{g, b\})$.

early results

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late results

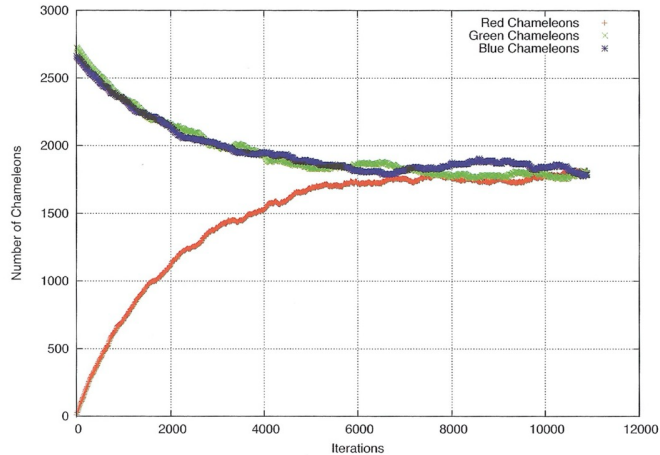


Figure 2.5 The first two generations (equivalent to 10,800 iterations) of an explicit simulation of encounters between chameleons changing colors. A total of 5,400 chameleons is simulated, starting from a condition where only blue and read chameleons are present, in virtually equal numbers, due to randomized initial condition. Red chameleons quickly pick up in numbers, until only fluctuations are left between numbers.

Example 2 ([2]). A soup of natural numbers divides each other...?

Let $\mathcal{X} = \mathbb{N} \setminus \{0, 1\}$.

Let $\mathcal{R} = \{x_1 + x_2 \rightarrow x_1 + x_3 : x_1, x_2, x_3 \in \mathcal{X} \wedge x_1 < x_2 \wedge x_3 = x_2/x_1\}$

Let $A(X) \sim \mathcal{A}(X, \mathcal{R})$.

Let $X_0 \sim \wp^*(\mathcal{X})$.



Experiment Time!

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Let $A(X) \sim \mathcal{A}(X, \mathcal{R})$.

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$$2+8 \rightarrow 2+4$$

$$3+21 \rightarrow 3+7$$

$\vdots$

$$2+3 \not\rightarrow ?$$

results?

2 

3 0

4 0

5 

6 0

7 

13

 mostly prime numbers

Example 3. A soup of single-digit numbers multiplies each other...?

Let $\mathcal{X} = \{0, 1, 2, 3, 4, 5, 6, 7, 8, 9\}$.

Let $\mathcal{R} = \{x_1 + x_2 \rightarrow x_3 + x_3 : x_1, x_2, x_3 \in \mathcal{X} \wedge x_3 = x_1 \cdot x_2 \bmod 10\}$

Let $A(X) \sim \mathcal{A}(X, \mathcal{R})$.

Let $X_0 \sim \wp^*(\mathcal{X})$.

$$2 + 8 \rightarrow 6 + 6$$

Experiment Time!

Example 3. A soup of single-digit numbers multiplies each other...?

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Let $A(X) \sim \mathcal{A}(X, \mathcal{R})$.

Let $X_0 \sim \wp^*(\mathcal{X})$.

results?

→ 0

